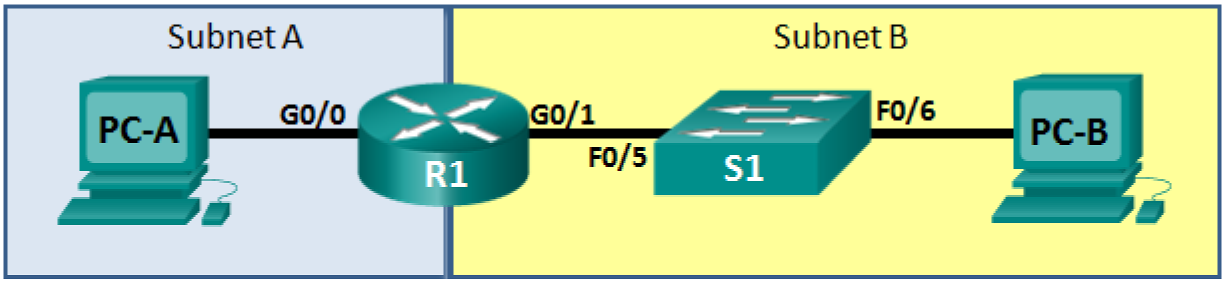
CCNA: Introduction to Networks 2024

Skills Assessment – Practice Exam

Topology



Assessment Objectives

Part 1: Develop the IPv4 Address Scheme

Part 2: Configure Device IPv4 and Security Settings

Part 3: Test and Verify IPv4 End-to-End Connectivity

Part 4: Configure IPv6 Addressing on R1

Part 5: Test and Verify IPv6 End-to-End Connectivity

Part 6: Cleanup

Scenario

In this Skills Assessment (SA) you will configure the devices in a small network. You must configure and PCs to support both IPv4 and IPv6 connectivity. You will configure security on the router. You will test and document the network using common CLI commands. Required Resources

- 1 Router (Cisco 1941 with Cisco IOS Release 15.2(4)M3 universal image or comparable)
- 1 Switch (Cisco 2960 with Cisco IOS Release 15.0(2) lanbasek9 image or comparable)
- 2 PCs (Windows 7, Vista, or XP with terminal emulation program, such as Tera Term)
- Console cable to configure the Cisco IOS devices via the console ports
- Ethernet cables as shown in the topology

Part 1: Develop the IPv4 Addressing Scheme

Given an IP address and mask of _192.168.25.0_/26__________________ (address / mask), design an IP addressing scheme that satisfies the following requirements. **NOTE:** You should use fixed length subnetting for all subnets.

Subnet	Number of Hosts
Subnet A	15 hosts
Subnet B	12 hosts

Note consider other variations for practice e.g different starting mask lengths and creating a different number of networks

No calculators may be used. Please fill in the following tables based upon the outcome of your subnetting calculations.

Specification	Student Input
New Subnet Mask (decimal)	
Maximum number of usable subnets	
Number of usable hosts per subnet	

Fill in the following table with the subnet information, for the subnets required. Not all rows in the table will be used.

Subnet Number	Network Address	First Usable Host Address	Last Usable Host Address	Broadcast Address
A				
B				
C				
D				
E				

Host computers will use the first IP address in the subnet. The network router will use the LAST network host address. The switch will use the second to the last network host address.

Write down the IP address information for each device:

Device	IP address	Subnet Mask	Gateway	Points
PC-A				
R1-G0/0			N/A	
R1-G0/1			N/A	
S1			N/A	
PC-B				

Before proceeding, verify your IP addresses with the instructor.

Instructor Sign-off Part 1: ______________________

Part 2: Configure Device IPv4 and Security Settings

Step 1: Configure host computers.

After configuring each host computer, record the host network settings with the **ipconfig /all** command.

PC-A Network Configuration		Points
Physical Address		
IP Address		
Subnet Mask		
Default Gateway		

PC-B Network Configuration		Points
Physical Address		
IP Address		
Subnet Mask		
Default Gateway		

Step 2: Configure R1.

Configuration tasks for R1 include the following:

Task	Specification	Points
Disable DNS lookup		
Router name	R1	
Domain name	ccna-lab.com	
Encrypted privileged exec password	ciscoenpass	
Console access password	ciscoconpass	
Telnet access password	ciscovtypass	
Block login attempts for 100 seconds if 3 failed attempts occur within 30 seconds		
Create an administrative user in the local database	Username: admin Password: class	
Set login on VTY lines to use local database		
Set the minimum length for passwords	10 characters	
Set VTY lines to accept ssh connections only		
Encrypt the clear text passwords		
MOTD Banner		
Interface G0/0	Set the description Set the Layer 3 IPv4 address Activate Interface	
Interface G0/1	Set the description Set the Layer 3 IPv4 address Activate Interface	
Generate a RSA crypto key	1024 bits modulus	

Step 3: Configure S1

Task	Specification	Points
Disable DNS lookup		
Host name	S1	
Telnet access password	cisco	
Interface Vlan1	Set the description Set the Layer 3 IPv4 address Activate Interface	
Default Gateway		

Part 3: Test and Verify IPv4 End-to-End Connectivity

Step 1: Verify network connectivity.

Use the ping command to test connectivity between all network devices.

Note: If pings to host computers fail, temporarily disable the computer firewall and retest. To disable a Windows 7 firewall, select Start > Control Panel > System and Security > Windows Firewall > Turn Windows Firewall on or off, select **Turn off Windows Firewall**, and click **OK**.

Use the following table to methodically verify connectivity with each network device. Take corrective action to establish connectivity if a test fails:

From	To	IP Address	Ping Results	Points
PC-A	R1, G0/0			
PC-A	R1, G0/1			
PC-A	PC-B			
PC-B	R1, G0/1			
PC-B	R1, G0/0			

In addition to the ping command, what other command is useful in displaying network delay and breaks in the path to the destination?

Step 2: Test & confirm SSH operation

Open up an SSH client and demonstrate to the instructor successfully establishing an SSH session with the Router.

Part 4: Configure IPv6 Addressing on R1

Given an IPv6 network address of **2001:DB8:ACAD::/48**, configure IPv6 addresses for the Gigabit interfaces on R1. Use **FE80::1** as the link-local address on both interfaces.

Step 1: Configure R1.

Configuration tasks for R1 include the following:

Task	Specification	Points
Configure G0/0 to use the first address in subnet A.	Assign the IPv6 unicast address Assign the IPv6 link-local address	
Configure G0/1 to use the first address in subnet B.	Assign the IPv6 unicast address Assign the IPv6 link-local address	
Enable IPv6 unicast routing.		

Part 5: Test and Verify IPv6 End-to-End Connectivity

Step 1: Obtain the IPv6 address assigned to host PCs.

PC-A IPv6 Network Configuration		Points
Physical Address		
IPv6 Address		
Default Gateway		

PC-B IPv6 Network Configuration		Points
Physical Address		
IPv6 Address		
IPv6 Default Gateway		

Step 2: Use the ping command to verify network connectivity.

IPv6 network connectivity can be verified with the ping command. Use the following table to methodically verify connectivity with each network device. Take corrective action to establish connectivity if a test fails:

From	To	IPv6 Address	Ping Results	Points
PC-A	R1, G0/0			
PC-A	R1, G0/1			
PC-A	PC-B			
PC-B	R1, G0/1			
PC-B	R1, G0/0			

Part 6: Cleanup

Unless directed otherwise by the instructor, restore host computer network connectivity, and then turn off power to the host computers.

Before turning off power to the router and switch, remove the NVRAM configuration files (if saved) from both devices.

Disconnect and neatly put away all LAN cables that were used in the Final.